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Structure and Function of Word Laterality in Early Childhood

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Many of the brain's functional regions display lateralization, meaning they are more specialized to one hemisphere than the other. The visual word form area (VWFA) is one such region; its specialization for written scripts, which develops only after literacy, is consistently dominant on the left in adults. The experience-dependent nature of the VWFA makes it a prime opportunity to investigate the development of functional laterality in the human brain. Therefore, we ask, "What neural correlates relate to the development of word laterality in children?" We tested two potential sources of this word laterality: (a) structural connectivity (s-connectivity) of the VWFA with the high-level language network in frontal and temporal cortices and (b) activation in these language regions. To do so, we scanned reading children (4–13 years) on a visual functional magnetic resonance imaging localizer to define the VWFA and its word selectivity and a separate auditory functional magnetic resonance imaging localizer to define language regions and their linguistic selectivity, and we collected diffusion-weighted imaging to examine white matter connectivity. Probabilistic tractography was used to quantify white matter connections between our subject-specific functional regions. VWFA word laterality was predicted by the positive interaction between laterality of s-connectivity to temporal language regions and laterality of temporal language regions' selectivity, suggesting that the VWFA tends to be more left-lateralized when both temporal language areas are left-lateralized and the VWFA has strong s-connectivity to that left-lateralized language network. More generally, our findings suggest that both s-connectivity to related regions and activation patterns in those connected regions may play a role in the development of functional laterality, and shed light on the development of the neural circuitry that underlies reading.

Keywords: visual word form area, laterality, language, connectivity, diffusion-weighted imaging

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How does our brain support the ability to read? The visual word form area (VWFA) is a region in the ventral temporal cortex that displays selectivity to written words, only emerges with reading acquisition (Baker et al., 2007; Dehaene et al., 2015), and is

consistently localized to the left hemisphere (LH; Cohen et al., 2000). Language processing is also typically dominant in the LH (Labache et al., 2020, 2023; Ojemann, 1991); does this language lateralization drive the development of laterality for orthographic

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processing (Cohen et al., 2000)? Better understanding the development of this component of cortical organization will further elucidate the high-level mechanisms that govern the development of human brain organization and provide insight into the factors that underlie individual variability in cortical organization.

Many neuroimaging studies have investigated potential relationships with the development of word laterality. One hypothesis is that the LH bias for high spatial frequency (HSF) visual information may support and serve as a precursor for the localization of words to the left (Ossowski & Behrmann, 2015). In contrast with this hypothesis, Ossowski and Behrmann (2015) did not find a significant LH bias for HSF information in young children prior to the acquisition of orthographic competence. However, they did find a correlation where children with relatively stronger development of a LH bias for orthographic processing had also developed a stronger LH bias for HSF information, implying that LH bias for HSF information may actually co-occur with or result from the development of orthographic processing. Another possibility is that it is the connections of visual cortex with language processing (in frontal and temporal areas) that drives the development of a lateralized VWFA. This idea builds off the more general connectivity hypothesis (Passingham et al., 2002), which suggests that a region's functional specialization is driven by its pattern of connectivity. Several previous studies have examined the connectivity of the VWFA in literate adults. The mature VWFA displays preferential resting state connectivity with language areas (Stevens et al., 2017). Further, Chen et al. (2019) found that both structural and functional connectivity analyses reveal patterns of connectivity between the VWFA and areas of the language network in adults and that this structural connectivity predicts language and reading ability. While this prior work examines the connectivity of the mature VWFA, a study by Saygin et al. (2016) demonstrated that the connectivity hypothesis is developmentally applicable with their finding that the spatial profile of a child's VWFA at age 8 could be reliably predicted by the child's structural connectivity fingerprint at age 5, before they were reading. Later work further demonstrated that the putative VWFA of neonates already shows stronger connectivity with language regions than adjacent regions, which suggests VWFA–language connectivity may play an important role in the VWFA's localization and development (Li et al., 2020). Further, the language network functionally emerges prior to the VWFA (Dehaene-Lambertz et al., 2006), enabling connectivity with language to be investigated as a factor that may predispose the cortical region that becomes the VWFA to develop word selectivity during reading acquisition, especially since both functions are left-lateralized (Cohen et al., 2000; Ojemann, 1991).

No previous studies have explored the relationship between structural connectivity to language regions and VWFA functional laterality in reading children. Therefore, in this study, we use subject-specific functional magnetic resonance imaging (fMRI) to define both visual word and language processing centers in each child's brain to investigate neural correlates of word laterality in young children. We ask, does structural connectivity to high-level language regions relate to laterality of word selectivity in children? In line with the connectivity hypothesis and existing literature, we hypothesize that stronger leftward asymmetry of VWFA to language s-connectivity will predispose the VWFA to develop stronger LH dominance of word selectivity.

Additionally, we test the interactive specialization hypothesis (which is not mutually exclusive with the connectivity hypothesis) that states that a region's functional specialization relates to both its

connectivity patterns and the activity patterns in the regions it is connected to (Johnson et al., 2009). As already discussed, the VWFA displays functional and structural connectivity with language regions (Chen et al., 2019; Li et al., 2020; Stevens et al., 2017), which are also left-lateralized (Ojemann, 1991). Gerrits et al. (2019) found that in adults, the laterality index for language (based on a word production task) was significantly and positively associated with the laterality index for word recognition (derived from a visual localizer task). However, no existing studies investigate potential relationships between word and language selectivity or laterality in children, leaving open the question of whether a relationship between higher level language and word specialization exists during development. Given this knowledge gap, we ask the following: Do young children show a relationship between language and word selectivity or laterality? We hypothesize that VWFA word laterality will be related to language selectivity in high-level language regions.

We use fMRI to establish subject-specific functional regions of interest (fROIs) for the VWFA and language regions. We then extract selectivity from these regions and calculate laterality. To estimate structural connectivity between our subjects' individual fROIs, we use probabilistic tractography with subject-specific fROIs as seeds and targets. Our results offer a clearer picture of the development of lateralization in the VWFA and the neural substrates of reading behavior.

Results

Word Selectivity Is Lateralized in Young Children

First, we asked whether our sample of young children displays left-lateralized word selectivity. We compared mean word selectivity in the LH VWFA with mean word selectivity in the right hemisphere VWFA via a paired-samples *t* test. As expected, word selectivity was significantly stronger in the left VWFA than in the right VWFA ($p = 7.24 \times 10^{-11}$, $n = 72$; Figure 1), justifying further analyses with magnitude of VWFA laterality.

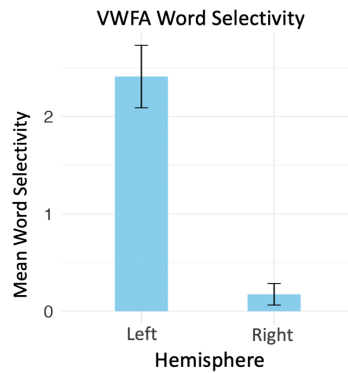
Laterality of Word Selectivity Does Not Increase With Age

Next, we asked if word laterality increased with age. A linear mixed effects model between VWFA left laterality and age revealed that the relationship is nonsignificant ($\beta = .22$, $p = .14$, $n = 72$; Figure 2), suggesting that other factors may drive variability in word laterality during development.

VWFA Laterality Predicted by Positive Interaction Between Laterality of Temporal Language Regions and Laterality of S-Connectivity to Temporal Language Regions

Next, we asked whether, in line with the connectivity hypothesis, the magnitude of VWFA word laterality relates to its structural connectivity with subject-specific language regions defined via an additional fMRI task. We also sought to test our hypothesis that language selectivity in these frontotemporal language regions would be related to VWFA laterality, as suggested by the interactive specialization hypothesis. For both s-connectivity to language and

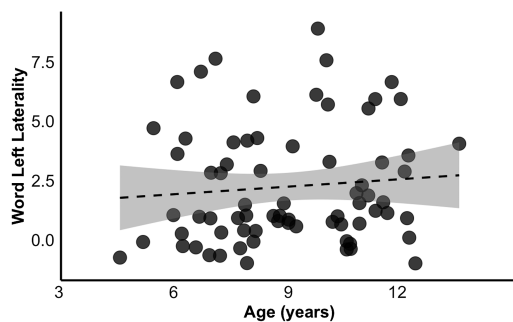
Figure 1
Word Selectivity Being Significantly Stronger in the Left Hemisphere VWFA



Note. Word selectivity is significantly stronger in the left VWFA than the right VWFA. Mean word selectivity is plotted for subject-specific VWFA defined on the left and on the right hemisphere. Lines indicate standard error across participants. VWFA = visual word form area. See the online article for the color version of this figure.

laterality of language selectivity, we computed s-connectivity and/or laterality averaged across all frontal ROIs and across all temporal ROIs separately. To capture relative weighting and interactions between these variables of interest, we ran a mixed effects linear model with VWFA laterality predicted by frontal language laterality of selectivity, temporal language laterality of selectivity, laterality of s-connectivity to frontal language, laterality of s-connectivity to temporal language regions, the interaction between frontal language laterality and laterality of s-connectivity to frontal language, and the interaction between the laterality of temporal language regions and laterality of their s-connectivity, while including subject identity as a random effect. We included age as a covariate to ensure any

Figure 2
Word Laterality Being Not Significantly Associated With Age



Note. Visual word form area word laterality is not significantly related to age. Left lateralization for word selectivity (left visual word form area's word selectivity–right visual word form area's word selectivity) is plotted for each participant, with the dashed line indicating linear fit and the shaded region representing the 95% confidence interval.

apparent relationships were not primarily driven by age effects. The strongest and only significant predictor of VWFA word laterality was the positive interaction between laterality of temporal language regions and the laterality of their s-connectivity with VWFA ($\beta = .62, p = .015, n = 69$; Figure 3). No other main effects or interactions were significant (Figure 4; frontal language laterality by s-connectivity: $\beta = -.012, p = .96, n = 69$; main effects of frontal language laterality: $\beta = .30, p = .38, n = 69$; laterality of temporal language regions: $\beta = .34, p = .22, n = 69$; laterality of frontal s-connectivity: $\beta = .19, p = .50, n = 69$; laterality of temporal s-connectivity: $\beta = -.11, p = .72, n = 69$).

Reading Scores Predicted by Word Left Laterality and Interaction Between Laterality of Temporal Language Selectivity and S-Connectivity

Next, we sought to investigate how VWFA word laterality, laterality of s-connectivity to language, and laterality of selectivity in language regions may be related to reading behavior. We ran a linear mixed effects model predicting reading scores (standardized scores on the Kaufman Test of Educational Achievement, Third Edition [KTEA-3]) from VWFA word laterality, laterality of s-connectivity to both frontal and temporal language regions, laterality of language selectivity in both frontal and temporal language regions, interactions between VWFA word laterality and each of the other terms, and interactions between laterality of s-connectivity and laterality of selectivity for both frontal and temporal language areas. Word left laterality was a significant predictor of reading scores ($\beta = 2.45, p = .025, n = 57$; Table 1, Figure 5). Additionally, the interaction between temporal language areas' laterality and laterality of s-connectivity to temporal language areas significantly predicted reading scores ($\beta = -4.89, p = .036, n = 57$; Table 1).

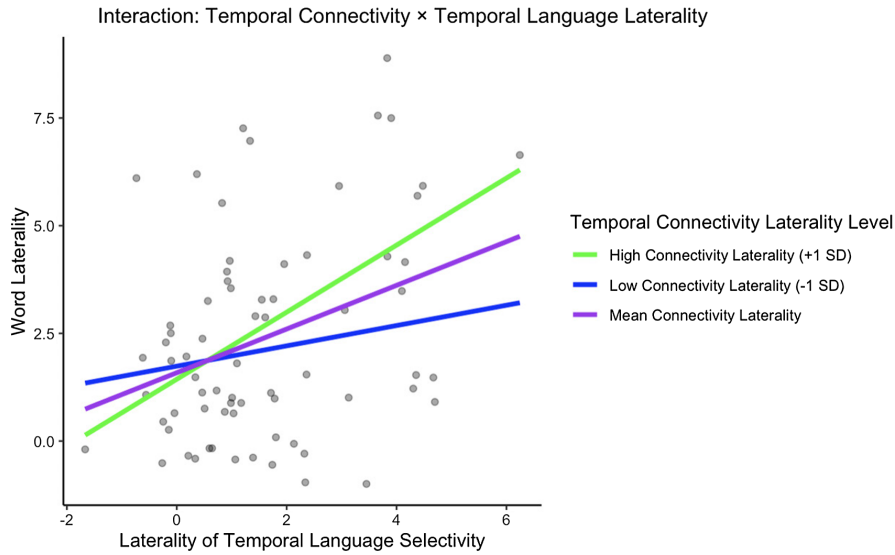
Discussion

Understanding the neural factors that relate to laterality of word selectivity during development will provide insight into what domain-general mechanisms may drive the development of functional laterality throughout the human brain, and they will improve our understanding of factors that drive reading. In this study, we examined several possible correlates of VWFA word left laterality in a sample of reading children. We used subject-specific fROIs with two separate tasks to maximize precision given individual variability in the anatomical location of the VWFA and high-level language regions. Our quantitative metric for structural connectivity was unique in its focus on s-connectivity between these functionally defined subject-specific fROIs, allowing us to investigate s-connectivity between functional regions with a high degree of precision. To our knowledge, this is the first fMRI study to investigate relationships between word laterality, structural connectivity, and language laterality in children.

In line with both the connectivity hypothesis and the interactive specialization hypothesis, we found that VWFA word laterality is predicted by a positive interaction between laterality of s-connectivity to temporal language regions and laterality of selectivity in temporal language regions. VWFA laterality tends to be strongest in participants who display both greater left laterality of s-connectivity to temporal language areas and greater left laterality of selectivity in temporal language areas.

Figure 3

Word Laterality Predicted by Positive Interaction Between Temporal Language Regions' Laterality and Connectivity



Note. Visual word form area word laterality is predicted by a positive interaction between functional laterality of temporal language regions (to auditory language) and their s-connectivity with the visual word form area. The left lateralization of temporal language regions and left lateralization of visual word form area word selectivity are plotted for each participant. Colored lines indicate model predictions (word laterality $\sim$ laterality of s-connectivity to temporal language areas $\times$ laterality of selectivity of temporal language areas) for participants with high (green; 1 standard deviation above mean) and low (blue) laterality of temporal language s-connectivity; mean laterality of s-connectivity of temporal language regions plotted in purple. See the online article for the color version of this figure.

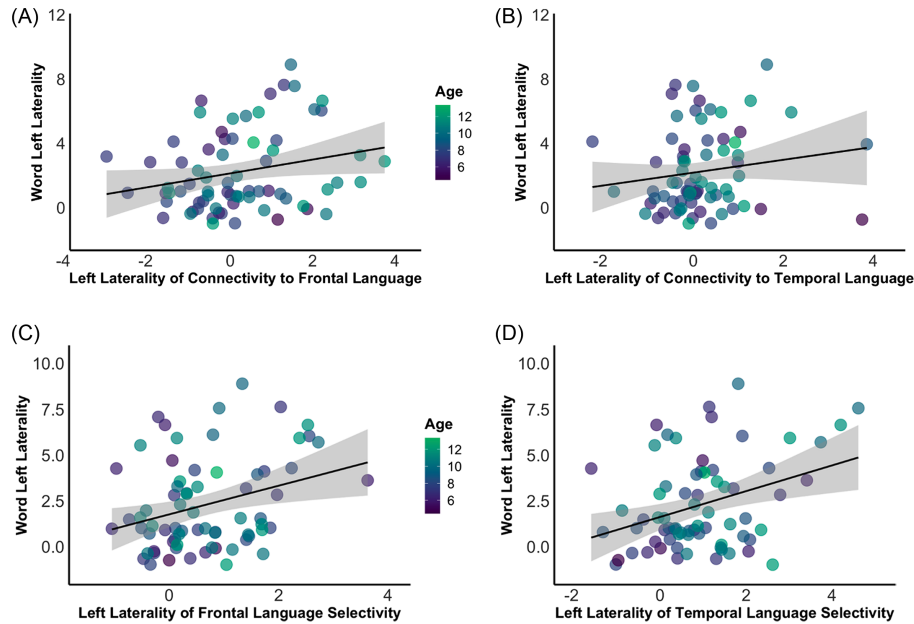
In other words, participants with stronger left laterality of s-connectivity to temporal language regions tend to have a stronger relationship between temporal language regions' laterality and VWFA word laterality. Conversely, participants with stronger laterality in temporal language regions tend to have stronger relationships between laterality of s-connectivity to temporal language areas and VWFA word laterality. Our model was age adjusted, so age does not appear to be the primary factor driving these relationships. While the applicability of the connectivity hypothesis in the development of VWFA localization has been previously demonstrated in prereaders (Saygin et al., 2016), our study is the first to test this hypothesis with respect to laterality in young readers. Our finding that VWFA word laterality tends to be strongest in individuals with both stronger lateralization of s-connectivity to temporal language areas and stronger lateralization of language selectivity provides support for the connectivity hypothesis's applicability to the development of word laterality. Our finding suggests that s-connectivity with the left-lateralized language network is important for the development of VWFA laterality, a notion that is also supported by previous work. For instance, lesion studies have found that patients with a right-lateralized language network due to LH damage during early childhood also tend to display right lateralization of word processing (Li et al., 2022; Seydell-Greenwald et al., 2025). Additionally, even in the absence of lesions, participants who have a dominant language network on the right also have a right-lateralized VWFA (Cai et al.,

2010; Gerrits et al., 2019; Seydell-Greenwald et al., 2025; Van der Haegen et al., 2012).

Additionally, our finding that VWFA laterality is predicted by a positive interaction between language s-connectivity and language laterality aligns with the interactive specialization hypothesis; our finding suggests that both s-connectivity and ongoing functional specialization of connected regions seem to drive word laterality in visual cortex. As language regions become increasingly selective for linguistic content and as children learn to read, language regions synchronize with the VWFA, leading to stronger s-connectivity and increased selectivity. Our support for the influence of language laterality in children is similar to previous findings in adults demonstrating a relationship between language laterality and laterality of visual word processing (Gerrits et al., 2019). It is notable that this relationship crosses functional domains and sensory modalities; we see that the magnitude of hemispheric specialization for the processing of *visual* words is related to the magnitude of hemispheric specialization for higher level linguistic processing of *auditory* language, including properties such as semantics and syntax.

Our finding that word left laterality and the interaction between laterality of temporal language areas and laterality of s-connectivity to temporal language areas predict reading scores in our sample suggests behavioral relevance of functional laterality and factors that relate to functional laterality, such as patterns of s-connectivity to

Figure 4
Word Laterality Plotted Against Language Regions' Laterality of Selectivity and Connectivity



Note. (A, B) Left lateralization for word selectivity (left visual word form area's [VWFA] word selectivity–right VWFA's word selectivity) and left lateralization of VWFA to language s-connectivity (left VWFA s-connectivity to left language regions–right VWFA s-connectivity to right language regions) are plotted for each participant, with the dashed line indicating linear fit and the shaded region representing the 95% confidence interval. Points are color-coded by participant age. (C, D) Left lateralization for word selectivity (in VWFA) and left lateralization for language selectivity (in language regions) are plotted for each participant, with the dashed line indicating linear fit and the shaded region representing the 95% confidence interval. Points are color-coded by participant age. See the online article for the color version of this figure.

related regions. However, the cross-sectional nature of this study means our findings are correlational, not causal, so we can only conclude that similar processes may influence both reading ability and laterality of orthographic processing during development. Future work should continue to explore behavioral relevance in longitudinal samples to elucidate whether functional laterality has causal relevance for behavior.

This study leaves open and drives several further questions. A limitation of the present study is its cross-sectional design; we cannot establish directionality or causation between variables. We can only conclude that VWFA laterality, s-connectivity to language, language laterality, and reading ability are influenced by similar factors or developmental processes. For instance, there may be a third factor that drives both orthographic processing and higher level language to develop LH dominance in most individuals, and this factor may also influence reading ability. However, it is possible, for example, that word laterality does increase across time within an individual and that s-connectivity (and/or selectivity) of language regions drives this increase in lateralization. A longitudinal design would allow us to test this hypothesis within individuals. Future studies can also measure within-subject gains in reading skills and relate these to gains in lateralization and s-connectivity with a longitudinal design. Finally, our findings suggest that the connectivity hypothesis and

interactive specialization hypothesis may be applicable to other lateralized functional domains. Future work could explore whether s-connectivity and activation in connected regions relate to magnitude of laterality in other lateralized functions such as face processing.

Overall, our study provides insight into the potential roles of connectivity and ongoing development and maturation of auditory language in the development of reading, which will be useful for future education and developmental research.

Method

Participants

Forty-seven healthy children with a total of 72 scanning sessions (ages 4.45–13.34, $M = 8.85$) were recruited from the Columbus and Ohio State University areas and completed structural scanning and functional scanning as part of several ongoing studies. All participants included in this study were reading (as defined by parent report and the KTEA-3 Letter and Word Recognition when available), which ensured the VWFA could be robustly defined in our sample given that word selectivity emerges with reading acquisition (Dehaene et al., 2015). No subjects included in our main analyses met exclusion criteria for high motion (see the Data Acquisition and Preprocessing section). All individuals reported no

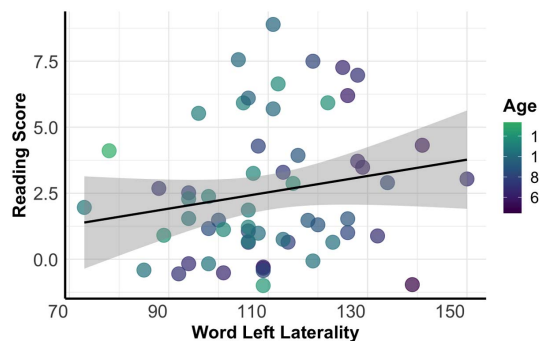
Table 1
Models Predicting Reading Scores

Term	Estimate	<i>p</i>
VWFA word left laterality	2.45	.025
Laterality of frontal language selectivity	2.61	.48
Laterality of temporal language selectivity	2.40	.39
Laterality of s-connectivity to frontal language regions	−2.20	.36
Laterality of s-connectivity to temporal language regions	.31	.91
VWFA Word Laterality × Laterality Frontal Language Selectivity	−.52	.57
VWFA Word Laterality × Laterality Temporal Language Selectivity	−.53	.31
VWFA Word Laterality × Laterality of S-Connectivity to Frontal Language Regions	.69	.32
VWFA Word Laterality × Laterality of S-Connectivity to Temporal Language Regions	.37	.68
Laterality Frontal Language Selectivity × Laterality S-Connectivity to Frontal Language Regions	−.22	.90
Laterality Temporal Language Selectivity × Laterality S-Connectivity to Temporal Language Regions	−4.89	.036

Note. The table shows the output from linear mixed effects model predicting reading scores from VWFA laterality, laterality of s-connectivity to frontal and temporal language regions, laterality of frontal and temporal language regions' selectivity, and interactions between these terms. Laterality of VWFA word left laterality and the interaction between laterality of s-connectivity to temporal language areas and laterality of temporal language selectivity reach significance. VWFA = visual word form area.

developmental, neuropsychological, or neurological conditions; had normal or corrected-to-normal vision and hearing; and were fluent English speakers. Parental consent and child assent were collected for all participants, and all participants were compensated for their time. The study received approval from the Institutional Review

Figure 5
Reading Scores Significantly Relating to Word Laterality



Note. Reading score and visual word form area word left laterality are plotted for each participant, with the dashed line indicating linear fit and the shaded region representing the 95% confidence interval. Points are color-coded by participant age. See the online article for the color version of this figure.

Board at the Ohio State University (IRB 2017H0387 and IRB 2021H0083).

Experimental Design

Each participant completed a full imaging battery including six functional localizers, resting state, T1- and T2-weighted images, and diffusion-weighted imaging (DWI) acquisition. The entire scanning procedure lasts approximately 90 min. For this study, only the static visual localizer data, auditory language localizer data, structural data, and diffusion data were processed. A subset of participants also participated in a behavioral assessment session, including administration of the KTEA-3, with an approximate total duration of 90 min (Kaufman & Kaufman, 2014).

Functional Localizer Tasks

VWFA Static Visual Localizer

To define the VWFA, we used a static visual localizer task (Saygin et al., 2016) during which participants were quickly presented with images from four categories: objects, faces, words, and scrambled words. The images were presented in blocks, each of which contained 26 stimuli of the same visual category including two repetitions that participants were asked to identify as part of a one-back task to maintain attention. Refer to Saygin et al. (2016) for further task details.

Auditory Language Task

Regions of the high-level language network were defined using an auditory language task (Fedorenko et al., 2010). Participants listened to blocks from three categories: meaningful English sentences (Sn); nonsense sentences (Ns), which controlled for prosody but were created from phonemically intact nonsense words; and texturized sound (Tx), which controlled for low-level auditory features. All participants completed two runs of the language localizer task during each scanning session, except for two participants who only completed one run. Refer to Fedorenko et al. (2010) for further task details.

Data Acquisition and Preprocessing

All magnetic resonance imaging images were collected using a Siemens Prisma 3T scanner with a 32-channel phase array receiver head coil. The scanner was located at the Ohio State University's Center for Cognitive and Behavioral Brain Imaging. A whole-head, high-resolution T1-weighted magnetization-prepared rapid acquisition with gradient echo scan was collected for every participant (repetition time (TR) = 2,300 ms, echo time (TE) = 2.9 ms, and voxel resolution = 1.00 mm³). The diffusion-weighted images were collected from 64 diffusion-weighted directions with 2 mm isotropic voxels (TR/TE = 4570/85, acceleration = 2, *b* = 0 and 2000).

The static visual localizer task, used to localize the VWFA, was collected using echo-planar imaging sequence (TR = 2,000 ms, TE = 30 ms). For some participants, 2 × 2 × 2 mm³ voxels with 100 × 100 base resolution were collected, resulting in 25 slices that were approximately parallel to the base of the temporal lobe in order to acquire coverage of the whole ventral temporal cortex. For the remaining participants, 2.2-mm isotropic voxels were collected to

obtain whole-brain coverage with 54 slices. The auditory language localizer was acquired using echo-planar imaging sequence (TR = 1,000 ms, TE = 28 ms, and voxel resolution = $2 \times 2 \times 3 \text{ mm}^3$). Base resolution was 120×120 , and 56 slices were collected for whole-brain coverage.

Structural magnetic resonance imaging data processing was conducted using a semiautomated processing stream: recon-all from FreeSurfer (Fischl, 2012). Significant steps of preprocessing included intensity correction, skull stripping, spatial smoothing, surface coregistration, cortical parcellation, and white matter and subcortical segmentation. DWI data preprocessing involved skull stripping the data (using Functional Magnetic Resonance Imaging of the Brain Software Library's brain extraction tool; Smith, 2002) prior to performing distortion correction via eddy (corrects for both subject motion and eddy currents), and finally fitting diffusion tensors to the eddy-corrected data. To prepare for probabilistic tractography using probtrackX, we also ran Functional Magnetic Resonance Imaging of the Brain Software Library's bedpostX, which outputs distributions of diffusion parameters for each voxel and models crossing fibers within every voxel (Bayesian estimation of diffusion parameters obtained using sampling techniques; Behrens et al., 2003, 2007). All DWI frames for every participant were subjected to a visual inspection to identify artifacts such as venetian blinding; one of the 72 scanning sessions met exclusion criteria (10+ frames with significant artifacts). Statistical analyses were rerun without this subject and can be found in Supplemental Table S1. To ensure data integrity and identify any participants with high motion, Tracks Constrained by Underlying Anatomy (Yendiki et al., 2011) was run on all participants, and any individual with any of Tracks Constrained by Underlying Anatomy's four motion summary metrics (Average Translation, Average Rotation, Percent Bad Slices, Average Dropout Score) more than 3 SDs above the mean was planned to be excluded from all main analyses; however, no participants in our sample met our motion exclusion criteria.

For all functional tasks, preprocessing involved the following steps. Data were motion corrected within each scan, detrended, spatially smoothed (using full width at half maximum = 3 mm kernel for the static visual localizer and full width at half maximum = 4 mm for the auditory language localizer), and registered to each subject's anatomical space with bbregister from FreeSurfer (Greve & Fischl, 2009). For the static visual localizer only, field maps were acquired prior to task runs so that distortion correction could be completed due to the reduced signal-to-noise ratio in the ventral temporal cortex (Li et al., 2024).

First-Level Analyses

Preprocessed functional data were analyzed using a first-level generalized linear model. The experimental conditions for each functional task were included as explanatory variables, and six orthogonal motion estimates acquired during preprocessing were included as additional nuisance regressors. Further, a final nuisance regressor flagged all timepoints with greater than 1 mm total vector motion, and all subjects had less than 25% timepoints that met this threshold. For both functional tasks, the canonical hemodynamic response function (standard g function, $d = 2.25$, and $t = 1.25$) was convolved using a block design with a standard boxcar function (events on/off). The contrast maps used for further analysis were the Sn–Ns (from the auditory language localizer, for evaluating robust

selectivity properties of high-level language such as semantics and syntax; Fedorenko et al., 2010) and Words–All Other Stimuli (from the static visual localizer).

Defining fROIs

To account for intersubject variability in the location, size, and shape of high-level visual regions like the VWFA, we used the group-constrained subject-specific method (Fedorenko et al., 2010) to define all fROIs. Parcels, which encompass the spatial region that an fROI is typically located, defined in previous studies (the VWFA parcel was from Saygin et al., 2016, and language parcels were from Fedorenko et al., 2010, but using an updated version with 220 participants) were used as the search spaces for identifying each individual's high-level visual regions and language regions. These parcels were registered to the FreeSurfer combined volumetric and surface-based registration average-35 Montreal Neurological Institute standard template space brain if not already provided in that space, and then registered to every participant's individual anatomical brain using FreeSurfer's combined volumetric and surface-based registration (mri_cvs_register; Zöllei et al., 2010). Then, the contrast map from each subject's first-level generalized linear model was intersected with the parcel to create an fROI by selecting the 1,600 most significant voxels for the appropriate contrast from the first run of task data. This was done to create bilateral fROIs for the VWFA (using the contrast Words > All) and five language regions (Sn > Ns; posterior temporal, anterior temporal, inferior frontal, middle frontal, and orbital inferior frontal). Any voxel that was assigned to multiple fROIs was assigned to only the fROI for which it was most significant.

These fROIs were then registered to DWI space using Advanced Normalization Tools (Version 2.1.0, <http://stnava.github.io/ANTs>) and used for further structural connectivity analysis. The large initial fROI size of 1,600 voxels accounted for fROI shrinkage resulting from registration to DWI space. Once registered to the DWI space, these fROIs had a size of about 200 voxels. An additional set of fROIs was created using only the 150 most selective voxels and used to calculate selectivity and laterality. For each individual, a 150-voxel fROI was created using the first run of task data, and then another fROI was created using the second run of the task data. This was done so that selectivity could be calculated from an independent set of data for each fROI, but also allowed averaging of selectivity across the two orders for each participant. The two participants with only one run of the language task were not used in analyses that required two runs of data (i.e., they did not have language selectivity and laterality measures). Further, VWFA fROIs in these two participants with only one run of the language task were defined independently of language fROIs; therefore, some overlap may exist between temporal language regions and ventral visual regions, though this is unlikely given that their respective parcels are primarily nonoverlapping.

Calculating Selectivity

For our measure of selectivity, we took the average of all voxels' t -statistics for each of the two contrasts within the previously defined 150 voxel fROIs using an independent run of data from the run used to define the fROI. We then averaged these selectivity values across both run orders (the average for defining fROIs with Run 1 and

extracting selectivity with Run 2 and defining fROIs with Run 2 and extracting selectivity with Run 1).

Calculating Laterality

For all fROIs, laterality was calculated by subtracting the right hemisphere fROI's selectivity (as defined above) for the preferred contrast from the LH fROI's selectivity (i.e., LH selectivity – RH selectivity). For example, word left laterality for each subject was calculated by subtracting their right VWFA's selectivity for the W-All contrast from their left VWFA's selectivity for the W-All contrast. Therefore, positive laterality values indicate left laterality while negative values indicate right laterality.

Estimating Structural Connectivity Between fROIs

To estimate structural connectivity between our subject-specific fROIs, we used Functional Magnetic Resonance Imaging of the Brain Software Library's probtrackX Version 2 and treated each fROI as a seed and all other fROIs as targets. We set 10,000 samples per seed voxel and kept all other parameters at default (step length = .5 mm, max steps per streamline = 2,000, curvature threshold = .2) and used each participant's skull-stripped brain as a mask. Our metric for asymmetrical s-connectivity between each pair of fROIs was a ratio expressing the proportion of generated streamlines from the seed fROI that reached the specific target fROI (average across all voxels of how many streamlines from each voxel of the seed fROI reached the target fROI divided by waytotal). For all statistical analyses, this asymmetric s-connectivity metric was averaged in both directions (e.g., averaging this value when the VWFA is the seed and the middle frontal language fROI is the target with the middle frontal language fROI as the seed and the VWFA as the target) to create a symmetrical measure of s-connectivity between the pairs of fROIs. Any symmetrical s-connectivity values of 0 were assigned NaN, and then each symmetrical s-connectivity value was log-transformed to generate normally distributed values and account for the clustering of values near 0.

Reading Scores

Reading ability was assessed using the KTEA-3 standardized reading score. KTEA-3 reading scores were collected at 57 of our 72 scanning sessions.

Statistical Analyses

Linear mixed effects models were used to evaluate relationships between word laterality and our other variables of interest (a model predicting VWFA laterality from age and a separate combined model predicting VWFA laterality from laterality of VWFA s-connectivity to both frontal and temporal language regions, left laterality of both frontal and temporal language regions' selectivity, and interactions between laterality of s-connectivity to language and laterality of language selectivity in both frontal and temporal cortices) while accounting for age as a fixed effect and subject identity as a random effect. The equation for the linear mixed effects model predicting word left laterality from age is $\text{word_left_laterality} = \text{Age} + (1|\text{personID})$. The equation for the combined linear mixed effects model predicting word left laterality is $\text{word_left_laterality} = \text{Laterality_Frontal_Language_}$

$\text{Connectivity} \times \text{Laterality_Frontal_Language_Selectivity} + \text{Laterality_Temporal_Language_Connectivity} \times \text{Laterality_Temporal_Language_Selectivity} + \text{Age} + (1|\text{personID})$. A combined model was also run to predict KTEA-3 standardized reading scores from VWFA laterality, laterality of VWFA s-connectivity to both frontal and temporal language regions, left laterality of both frontal and temporal language selectivity, interactions between VWFA word laterality and these other variables of interest, and interactions between laterality of s-connectivity to language and laterality of language selectivity in both frontal and temporal cortices) while accounting for subject identity as a random effect. The equation for this model is $\text{KTEA} = \text{word_left_laterality} \times \text{Laterality_Frontal_Language_Connectivity} + \text{word_left_laterality} \times \text{Laterality_Temporal_Language_Connectivity} + \text{word_left_laterality} \times \text{Laterality_Frontal_Language_Selectivity} + \text{word_left_laterality} \times \text{Laterality_Temporal_Language_Selectivity} + \text{Laterality_Frontal_Language_Connectivity} \times \text{Laterality_Frontal_Language_Selectivity} + \text{Laterality_Temporal_Language_Connectivity} \times \text{Laterality_Temporal_Language_Selectivity} + \text{Age} + (1|\text{personID})$. Linear mixed effects models were conducted using MATLAB's fitlme function. The paired-samples *t* test between left and right VWFA word selectivity to determine whether word selectivity was significantly left-lateralized in our sample was run using MATLAB's ttest function. Scatterplots were created in RStudio (V 2023.06.2 +561) using the ggplot2 package (Wickham, 2023).

Transparency and Openness

All preprocessed data and code for statistical analyses are available at zeynepsaygin.com/ZlabResources.html. All statistical analyses were conducted in MATLAB Version R2025B (The MathWorks & Inc., 2025), and all plots were created using RStudio Version 2-23.06.2+561 (Posit, 2023) and the package ggplot2 (Wickham, 2023). Sample sizes/analytical approaches were not predetermined, but rather previously collected data were utilized and appropriate analytical methods for the available data were selected. This study and its design were not preregistered.

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